

Multimedia Project

Image Classification: Cats vs Dogs

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1. Introduction

This project aims to build an Image Classification system that distinguishes between cat and dog images using machine learning techniques studied in lectures. The goal is to understand the full pipeline including preprocessing, feature extraction, training, evaluation, and performance comparison.

2. Dataset Description

Cats images: 200

Dogs images: 200

Total images: 400

All images are resized to 128x128 pixels to ensure uniform processing.

3. Image Preprocessing & Enhancement

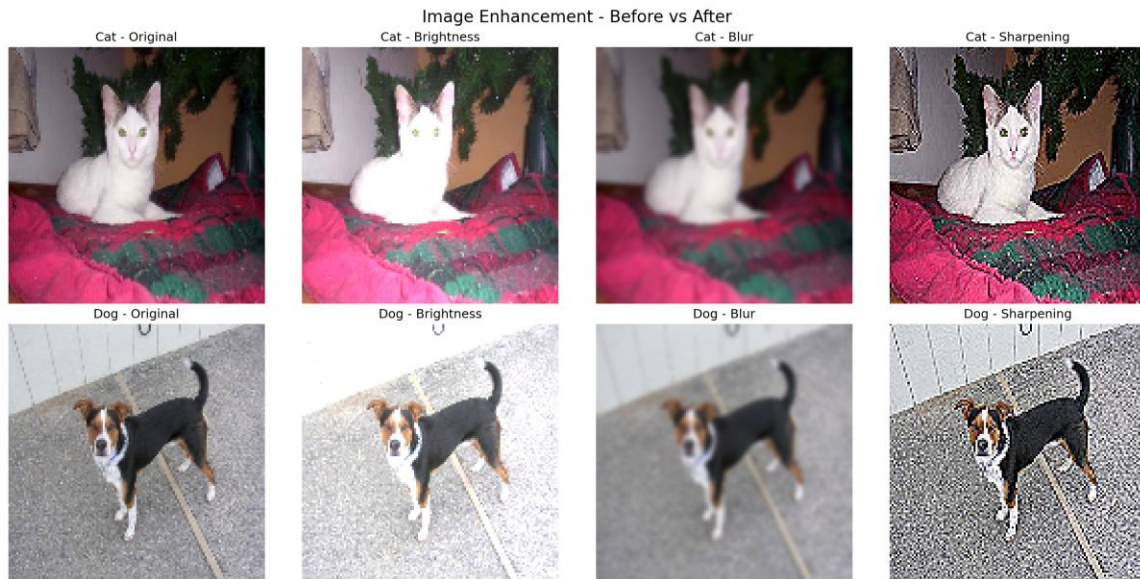
3.1 Techniques Used

- Image resizing (128x128)
- Brightness adjustment
- Gaussian blur
- Sharpening filter

3.2 Purpose

To improve image quality and reduce noise, which helps in better feature extraction.

3.3 Before / After Enhancement



3.4 Quantitative Impact of Image Enhancement

Model	Without Enhancement	With Enhancement	Improvement
KNN	54.00%	56.25%	+2.25%
SVM	56.25%	58.75%	+2.50%

Enhancement improved both models by reducing noise and improving feature quality.

4. Feature Extraction

- Histogram-based features used
- RGB color channels
- 16 bins per channel
- Total features per image = 48

Output shape: (400, 48)

5. Classification Models

5.1 KNN

- $k = 3$
- Distance-based classifier

5.2 SVM

- Kernel: RBF
- Works well for non-linear data

6. Train/Test Split

- Training set: 320 images
- Testing set: 80 images
- Stratified split used for balanced classes

7. Evaluation Metrics (Detailed)

7.1 Confusion Matrix Terms

Term	Meaning
TP	Dog correctly predicted as Dog
TN	Cat correctly predicted as Cat
FP	Cat predicted as Dog
FN	Dog predicted as Cat

7.2 Metrics Formulas

Accuracy = $(TP + TN) / \text{Total}$

Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

F1 = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

7.3 Numerical Example (SVM)

TP = 27

TN = 20

FP = 20

FN = 13

Results:

Accuracy = 58.75%

Precision = 57.45%

Recall = 67.50%

F1 = 62.07%

8. Confusion Matrix Results

KNN Confusion Matrix:

- True Negatives (Correct Cat): 24
- False Positives (Cat -> Dog): 16
- False Negatives (Dog -> Cat): 19
- True Positives (Correct Dog): 21

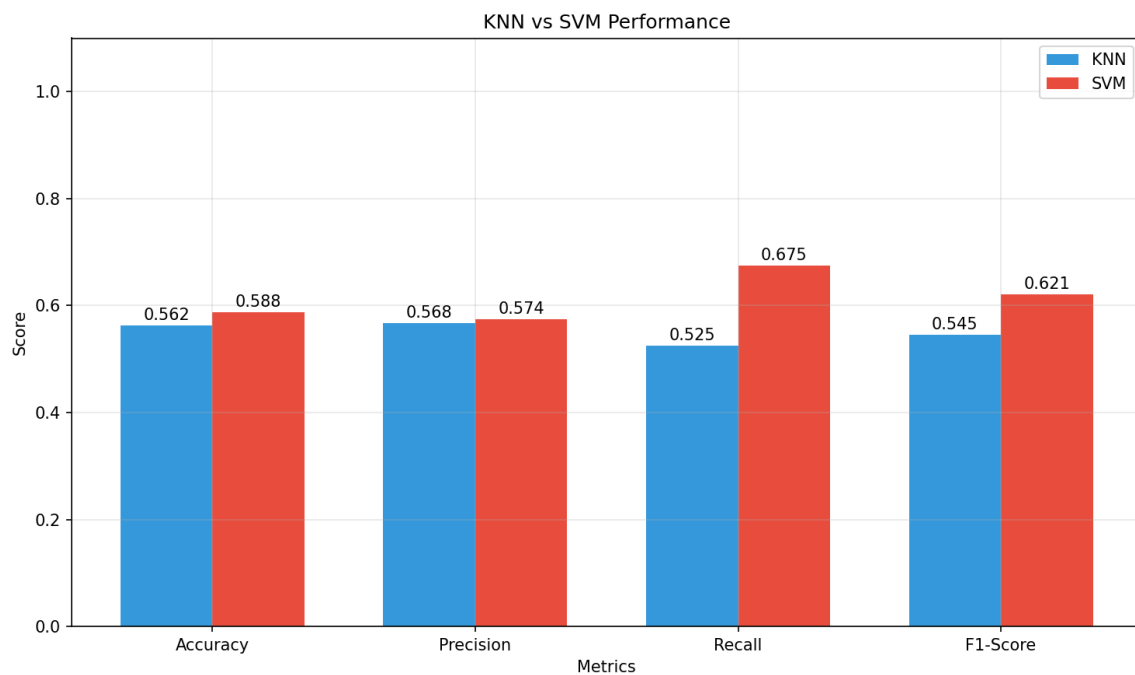
SVM Confusion Matrix:

- True Negatives (Correct Cat): 20
- False Positives (Cat -> Dog): 20
- False Negatives (Dog -> Cat): 13
- True Positives (Correct Dog): 27

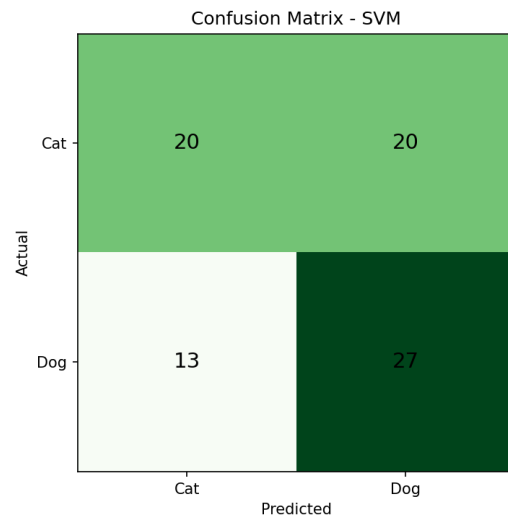
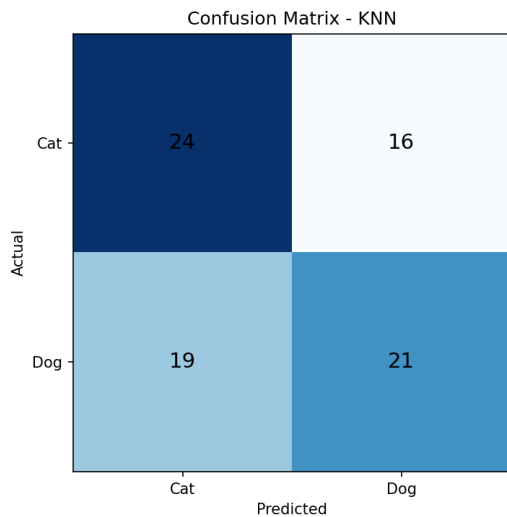
9. Model Comparison

Model	Accuracy	Precision	Recall	F1-Score
KNN	56.25%	56.76%	52.50%	54.55%
SVM	58.75%	57.45%	67.50%	62.07%

Best Model: SVM



10. Confusion Matrix Visualization



11. Effect of Image Enhancement

Technique	Effect
Brightness	Improves dark images
Sharpening	Enhances edges
Blur	Reduces noise

Overall improvement:

- KNN: +2.25%
- SVM: +2.50%

12. Error Analysis

Error Type	Reason
Similar colors	Histogram overlap
Background noise	Affects feature accuracy
Low lighting	Loss of visual details
Unusual poses	Feature variation

SVM Specific Errors:

- 13 dogs misclassified as cats (FN) -> difficulty detecting some dogs
- 20 cats misclassified as dogs (FP) -> false alarms

13. Conclusion

- SVM outperformed KNN
- Image enhancement improved performance
- Histogram features are simple but effective

- SVM is more stable for image classification

14. Generated Files

- image_enhancement_results.png
- models_comparison.png
- confusion_matrices.png